Practice Questions

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Date: 11/20/2021 8:52:37 pm • Time spent: 06:59

Score: 80% Passing Score: 80%

Question 1: ✔ Correct

You want to make sure that a set of servers will only accept traffic for specific network services. You have verified that the servers are only running the necessary services, but you also want to make sure that the servers will not accept packets sent to those services.

Which tool should you use?

- ➡ Port scanner
- System logs
- IDS
- Packet sniffer
- IPS

EXPLANATION

Use a port scanner to check for open ports on a system or a firewall. Compare the list of opened ports with the list of ports allowed by your network design and security policy. Typically, a port is opened when a service starts or is configured on a device. Open ports for unused services expose the server to attacks directed towards that port.

Use a packet sniffer to examine packets on the network. With a packet sniffer, you can identify packets directed towards specific ports, but you won't be able to tell if those ports are open. Examine system logs to look for events that have happened on a system, which might include a service starting, but would not likely reflect open ports.

An intrusion detection system (IDS) is a special network device that can detect attacks and suspicious activity. A passive IDS monitors, logs, and detects security breaches, but takes no action to stop or prevent the attack. An active IDS (also called an intrusion protection system or IPS) performs the functions of an IDS, but can also react when security breaches occur.

REFERENCES

LabSim for Network Pro, Section 14.1.

Question 2: ✔ Correct

What security mechanism can be used to detect attacks originating on the internet or from within an internal trusted subnet?

- ➡ IDS
- Security alarm
- Firewall
- Biometric system

EXPLANATION

An IDS is a security mechanism that can detect attacks originating on the internet or from within an internal trusted subnet.

A firewall is only able to filter traffic crossing through its interfaces. Therefore, it is unlikely to detect attacks within trusted areas that are not being filtered. A security alarm is a form of physical intrusion detection. Therefore, it is not suited for detecting electronic or online attacks. Biometric systems are a form of authentication mechanism; they are not a detection mechanism.

REFERENCES

LabSim for Network Pro, Section 14.1.

Question 3: ✔ Correct

What actions can a typical passive intrusion detection system (IDS) take when it detects an attack? (Select two.)

- [x] An alert is generated and delivered via email, the console, or an SNMP trap.
- [ ] LAN-side clients are halted and removed from the domain.
- [ ] The IDS configuration is changed dynamically, and the source IP address is banned.
- [x] The IDS logs all pertinent data about the intrusion.

EXPLANATION

The main functions of a passive IDS are to log suspicious activity and generate alerts if the attack is deemed severe. Additional functionality can be achieved by using a more advanced type of IDS called an active IDS. An active IDS can automate responses that may include dynamic policy adjustment and reconfiguration of supporting network devices to block the offending traffic.

REFERENCES

LabSim for Network Pro, Section 14.1.

Question 4: ✔ Correct

Which of the following activities are considered passive in regards to the function of an intrusion detection system? (Select two.)

- [x] Listening to network traffic
- [ ] Disconnecting a port being used by a zombie
- [ ] Transmitting FIN or RES packets to an external host
- [x] Monitoring the audit trails on a server

EXPLANATION

Passive IDS is a form of IDS that takes no noticeable action on the network. Passive IDS systems are undetectable to intruders. Passive IDS systems can monitor audit trails or listen to network traffic in real time.

Active IDS functions are those that interact with the network and generate detectable events. Such events can include disconnecting ports or transmitting FIN or RES packets to attackers.

REFERENCES

LabSim for Network Pro, Section 14.1.

Question 5: ✕ Incorrect

An active IDS system often performs which of the following actions? (Select two.)

- [ ] Update filters to block suspect traffic.
- [x] ~~Trap and delay the intruder until the authorities arrive.~~
- [x] Perform reverse lookups to identify an intruder.
- [ ] Request a second logon test for users performing abnormal activities.

EXPLANATION

An active IDS performs behaviors that can be seen by anyone watching the network. Usually, these actions are necessary to block malicious activities or discover the identity of an intruder. Updating filters and performing reverse lookups are common behaviors for an active IDS.

No form of IDS requires users to perform a second logon based on questionable activities. There are some authentication systems, such as CHAP, that periodically re-authenticate, but this is done at random time intervals, and the action is not visible to the user. A solution that serves to trap and delay the intruder until the authorities arrive describes a man trap (a physical security mechanism). However, this definition could be stretched to include honey pots and padded cells (logical or technical security mechanisms often used in conjunction with an IDS).

REFERENCES

LabSim for Network Pro, Section 14.1.

Question 6: ✔ Correct

Which of the following is the most common detection method used by an IDS?

- Anomaly
- ➡ Signature
- Behavior
- Heuristic

EXPLANATION

Signature recognition, also referred to as pattern matching, dictionary recognition, or misuse detection (MD-IDS), looks for patterns in network traffic and compares them to known attack patterns called signatures. Signature recognition is the most common IDS recognition type.

Anomaly recognition, also referred to as behavior, heuristic, or statistical recognition, monitors traffic to define a standard activity pattern as normal. Clipping levels or thresholds are defined that identify deviations from the norm. When the threshold is reached, an alert is generated or an action is taken.

REFERENCES

LabSim for Network Pro, Section 14.1.

Question 7: ✔ Correct

You have just installed a new network-based IDS system that uses signature recognition. What should you do on a regular basis?

- ➡ Update the signature files.
- Check for backdoors.
- Generate a new baseline.
- Modify clipping levels.

EXPLANATION

Signature recognition, also referred to as pattern matching, dictionary recognition, or misuse detection (MD-IDS), looks for patterns in network traffic and compares them to known attack patterns called signatures. Signature-based recognition cannot detect unknown attacks; they can only detect attacks identified by published signature files. For this reason, it is important to update signature files on a regular basis.

Anomaly recognition, also referred to as behavior, heuristic, or statistical recognition, monitors traffic to define a standard activity pattern as normal. Clipping levels or thresholds are defined that identify deviations from the norm. When the threshold is reached, an alert is generated or an action is taken.

REFERENCES

LabSim for Network Pro, Section 14.1.

Question 8: ✔ Correct

Which of the following are security devices that perform stateful inspection of packet data, looking for patterns that indicate malicious code? (Select two.)

- [x] IDS
- [x] IPS
- [ ] ACL
- [ ] Firewall
- [ ] VPN

EXPLANATION

An intrusion detection system (IDS) and an intrusion prevention system (IPS) are devices that scan packet contents looking for patterns that match known malicious attacks. Signature files identify the patterns of all known attacks. When a packet matches the pattern indicated in the signature file, the packet can be dropped or an alert can be sent.

Firewalls use an access control list (ACL) to filter packets based on the packet header (not data) information. Firewalls can filter packets based on port, protocol, or IP address. A virtual private network (VPN) is an encrypted communication channel established between two entities to exchange data over an unsecured network.

REFERENCES

LabSim for Network Pro, Section 14.1.

Question 9: ✔ Correct

Properly configured passive IDS and system audit logs are an integral part of a comprehensive security plan. What step must be taken to ensure that the information is useful for maintaining a secure environment?

- All logs should be deleted and refreshed monthly.
- ➡ Periodic reviews must be conducted to detect malicious activity or policy violations.
- The accounting department must compress the logs on a quarterly basis.
- All files must be verified with the IDS checksum.

EXPLANATION

Audit logs are useless unless they are periodically reviewed. The frequency will vary based on the criticality of the system being monitored, but the logs must be reviewed on a scheduled basis by a knowledgeable member of the IT/Infosec team.

REFERENCES

LabSim for Network Pro, Section 14.1.

Question 10: ✔ Correct

You are concerned about attacks directed at your network firewall. You want to be able to identify attacks and be notified of attacks. In addition, you want the system to take immediate action when possible to stop or prevent the attack.

Which tool should you use?

- IDS
- Port scanner
- Packet sniffer
- IPS

EXPLANATION

Use an intrusion prevention system (IPS) to both detect and respond to attacks. An intrusion detection system (IDS) can detect attacks and send notifications, but cannot respond to attacks.

Use a port scanner to check for open ports on a system or a firewall. Use a packet sniffer to examine packets on the network.

REFERENCES

LabSim for Network Pro, Section 14.1.

Question 11: ✕ Incorrect

As a security precaution, you have implemented IPsec between any two devices on your network. IPsec provides encryption for traffic between devices.

You would like to implement a solution that can scan the contents of the encrypted traffic to prevent any malicious attacks.

Which solution should you implement?

- Port scanner
- ➡ Host-based IDS
- ~~Network-based IDS~~
- Protocol analyzer
- VPN concentrator

EXPLANATION

A host-based IDS is installed on a single host and monitors all traffic coming in to the host. A host-based IDS can analyze encrypted traffic because the host operating system decrypts that traffic as it is received.

A network-based IDS is a dedicated device installed on the network. It analyzes all traffic on the network. It cannot analyze encrypted traffic because the packet contents are encrypted so that only the recipient can read the packet contents.

A protocol analyzer examines packets on the network, but cannot look at the contents of encrypted packets. A port scanner probes a device to identify open protocol ports. A VPN concentrator is a device that is used to establish remote access VPN connections.

REFERENCES

LabSim for Network Pro, Section 14.1.

Question 12: ✔ Correct

You are concerned about protecting your network from network-based attacks from the internet. Specifically, you are concerned about zero day attacks (attacks that have not yet been identified or that do not have prescribed protections).

Which type of device should you use?

- Network-based firewall
- ➡ Anomaly-based IDS
- Host-based firewall
- Anti-virus scanner
- Signature-based IDS

EXPLANATION

An anomaly-based intrusion detection system (IDS) can recognize and respond to some unknown attacks. Signature recognition, also referred to as pattern matching or dictionary recognition, looks for patterns in network traffic and compares them to known attack patterns called signatures. Signature-based recognition cannot detect unknown attacks; they can only detect attacks identified by published signature files.

Anti-virus software is a form of signature-based IDS. A network-based firewall filters packets for a network, while a host-based firewall filters packets for a host. Firewalls are typically configured using access control lists that identify specific traffic that is allowed or denied.

REFERENCES

LabSim for Network Pro, Section 14.1.

Question 13: ✕ Incorrect

Creating fake resources such as honeypots, honeynets, and tarpits fulfills which of the following main intrusion detection and prevention goals? (Select two.)

- [x] Offers attackers a target that occupies their time and attention while distracting them from valid resources.
- [ ] Reveals information about an attacker's methods and gathers evidence for identification or prosecution purposes.
- [ ] Detect attacks that are unique to the services on valid system resources and monitor application activity.
- [ ] Entices attackers to reveal their IDS signatures, which can then be matched to known attack patterns.
- [ ] Detect anomalous behavior that varies from standard activity patterns, also referred to as heuristic recognition.
- [x] ~~Lures attackers into a non-critical network segment where their actions are passively monitored and logged, then shuns the attacker by simply dropping their connection.~~

EXPLANATION

By using honeypots, honeynets, and tarpits you can fulfill the following intrusion detection and protection goals:

- Attackers are offered targets that will occupy their time and attention, distracting them from valid resources.
- You can observe attackers and gather information about their attack methods or gather evidence for identification or prosecution purposes.

REFERENCES

LabSim for Network Pro, Section 14.1.

Question 14: ✔ Correct

What does a tarpit specifically do to detect and prevent intrusion into your network?

- Passively monitors and logs suspicious activity until it detects a known attack pattern, then shuns the intruder by dropping their connection.
- Entices intruders by displaying a vulnerability, configuration flow, or data that appears to be of value.
- Uses a packet sniffer to examine network traffic and identify known attack patterns, then locks the attacker's connection to prevent any further intrusion activities.
- ➡ Answers connection requests in such a way that the attacking computer is stuck for a period of time.

EXPLANATION

A tarpit (also called a sticky honeypot) is a honeypot that answers connection requests in such a way that the attacking computer is stuck for a period of time.

REFERENCES

LabSim for Network Pro, Section 14.1.

Question 15: ✔ Correct

If maintaining confidentiality is of the utmost importance to your organization, what is the best response when an intruder is detected on your network?

- Record audit trails about the intruder.
- Monitor the intruder's actions.
- Delay the intruder.
- ➡ Terminate the intruder's session.

EXPLANATION

If maintaining confidentiality is of the utmost importance to your organization, then disconnecting an intruder is the best response. Allowing an intruder to spend any additional time inside of your network once discovered can lead to further breaches of confidentiality.

Delaying, auditing, and monitoring become important responses if prosecution is important.

REFERENCES

LabSim for Network Pro, Section 14.1.

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